📘 PROJECT REPORT

📰 News Article Summarizer

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1. 📌 Introduction

In today's fast-paced digital world, users often prefer quick access to information rather than reading long news articles. The **News Article Summarizer** is a Python-based application that extracts and summarizes news content from a given URL using Natural Language Processing (NLP).

This project provides a simple and efficient way to understand the core idea of any news article within seconds.

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2. 🎯 Objectives

- To build an application that summarizes news articles automatically
- To extract article content using a URL
- To apply NLP techniques for text summarization
- To create a user-friendly graphical interface

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3. 🛠️ Technologies Used

Technology	Purpose
Python	Core development
Tkinter	GUI interface
Newspaper3k	Article extraction
NLTK	Text processing

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4. 🧠 Methodology

1. User enters a news article URL
2. Article is downloaded and parsed
3. Text is tokenized into sentences
4. Word frequency is calculated
5. Sentences are ranked
6. Top sentences are selected as summary
7. Results are displayed in GUI

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5. 🖥️ System Architecture

User Input (URL)
↓
Article Extraction
↓
Text Processing
↓
Sentence Scoring
↓
Summary Generation
↓
GUI Output

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6. 📂 Project Structure

```
news-summarizer/
│
├── main.py
├── README.md
```

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7. ⚙️ Working of the Application

- The user inputs a URL into the application
- On clicking **"Summarize"**, the system:
  - Extracts article text
  - Processes the content
  - Generates a summary
- The title and summary are displayed
- The "Clear" button resets the application

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8. 📸 Output Screenshots

🔹 8.1 Application Interface

👉 Description:
This screen shows the main interface where the user can enter the URL and interact with the application.

🔹 8.2 Generated Summary Output

👉 Description:
The system displays the article title and a concise summary generated using NLP.

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9. ⚠️ Limitations

- Requires internet connection
- Some websites may block extraction
- Basic extractive summarization
- Limited UI features

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12. 📊 Applications

- Quick news reading
- Students and researchers
- Content summarization tools
- Media analysis

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13. ✅ Conclusion

The News Article Summarizer demonstrates how NLP techniques can be applied to real-world problems. It reduces reading time and provides users with quick insights into news content through a simple GUI.

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14. 📚 References

- Python Official Documentation
- NLTK Library
- Newspaper3k Library
- NLP Concepts

👨‍💻 Author

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Developed as part of a BYOP (Bring Your Own Project) to explore Python, NLP, and GUI development.